EX NO:3 DATE:	INSTALL GOOGLE APP ENGINE AND CREATE HELLO WORLD

AIM

To install Google App Engine and create Hello World app and deploy the app in the google cloud platform.

PROCEDURE:

Step 1: Create a Google Cloud Project

1. Open Google Cloud Console: https://console.cloud.google.com
2. Click on the Project Dropdown → Select New Project.
3. Enter a project name and click Create.
4. Select the newly created project.

Step 2: Open Cloud Shell and Set Project

1. Click the Cloud Shell (>_) icon in the top-right corner.
2. Click Authorize when prompted.
3. Set the active project using: gcloud config set project YOUR_PROJECT_ID

Step 3: Create Application Directory and Install Flask

1. Create a new directory and navigate into it: mkdir hello-app , cd hello-app
2. Install Flask : pip3 install flask

Step 4: Create the Flask Application

1. Create a file named main.py: nano main.py

```
from flask import Flask
app = Flask(__name__)
@app.route('/')
def hello():
    return 'Hello, World!'
```

```
if __name__== '__main__':
    app.run(host='0.0.0.0', port=8080)
```

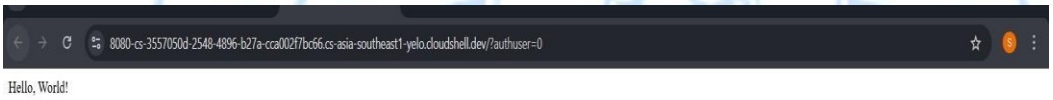
2. Save and exit the file.

Step 5: Execute and Verify the Application

1. Run the application: python3 main.py
2. Click Web Preview → Preview on port 8080 to view the output in the browser.
3. Output Displayed: 'Hello, World!'

OUTPUT:

```
C:\Users\kasab\AppData\Local\Google\Cloud SDK>gcloud projects list
PROJECT_ID                      NAME                PROJECT_NUMBER  ENVIRONMENT
gen-lang-client-0453894818      Gemini API          858980238712
hello-flask-project-487110      hello-flask-project 1078485799247
project-f1eccc3b-84a6-454e-823  My First Project    994599574269
```



RESULT:

Thus the implementation of Google App Engine and create Hello World app and deploy the app in the google cloud platform was executed and output verified successfully.

EX NO:4 DATE:	GAE LAUNCHER FOR WEB APPLICATION

AIM

To use Google App Engine (GAE) Launcher to deploy a web application.

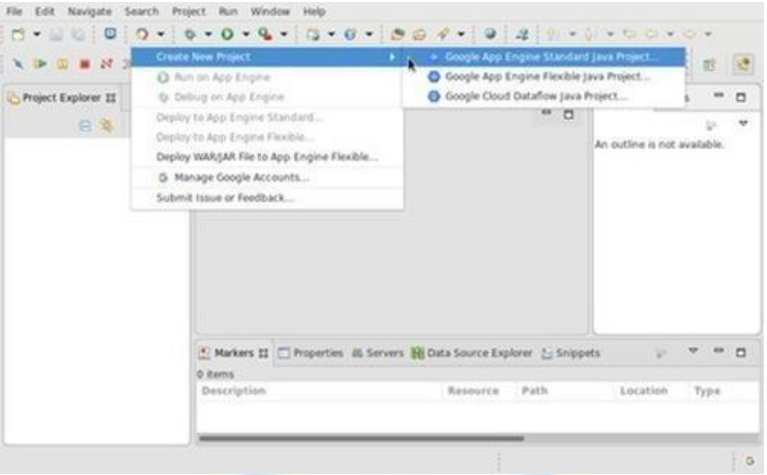
PROCEDURE:

Step 1: Download and Install GAE Launcher

1. Download the Google App Engine Launcher from the Google App Engine website (https://cloud.google.com/appengine/docs/standard) for your operating system.
2. Install the GAE Launcher by following the installation instructions provided.

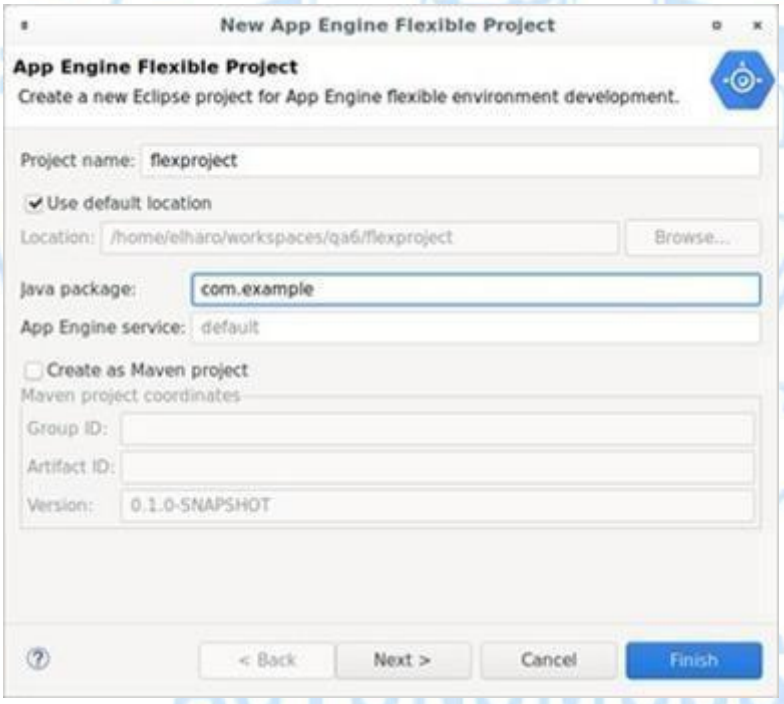
Step 2: Create a New Project

1. Open the GAE Launcher after installation.
2. Click on the "Create a new application" button.
3. Sign in with your Google account or create one if you don't have one.



Step 3: Create Application Directory

1. Enter a name for your project.
2. Choose your default runtime environment (Python or Java).
3. Select the location where you want to save your project files.
4. Click "Create" or "OK" to create your project.



Step 4: Create a Web Application

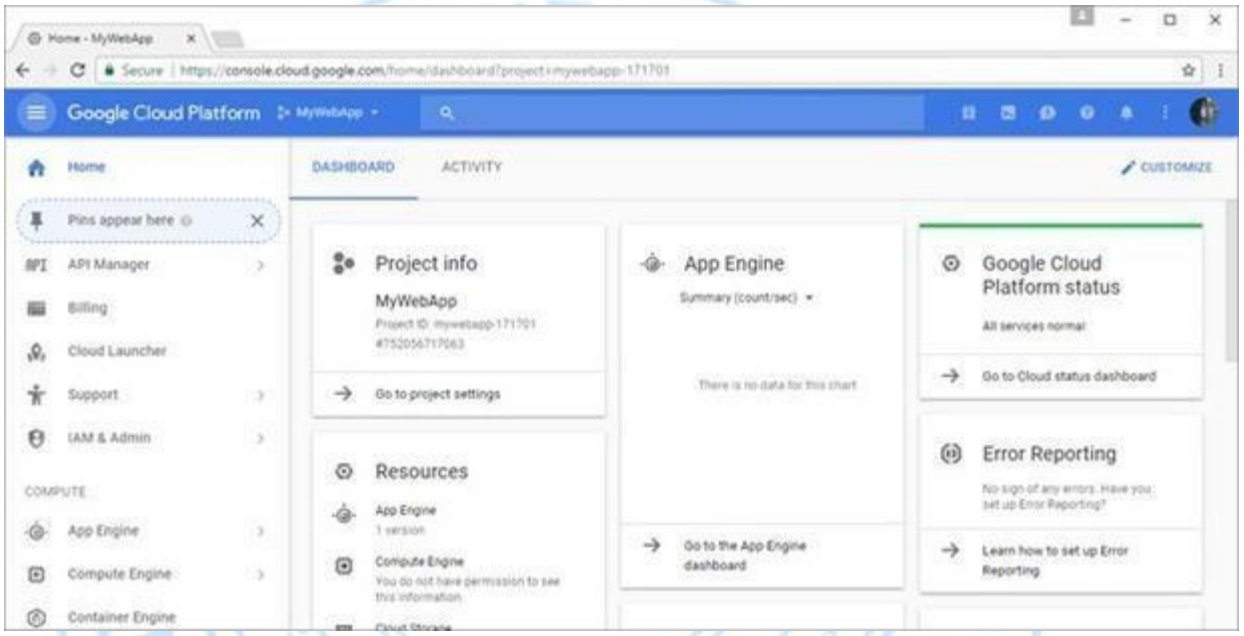
1. In the GAE Launcher, click the "Create an App Engine Web Application" button.
2. Choose a template (e.g., "Hello World") to create a sample web application.

Step 5: Edit and Test Your Web Application

1. The GAE Launcher will generate a sample web application for you.

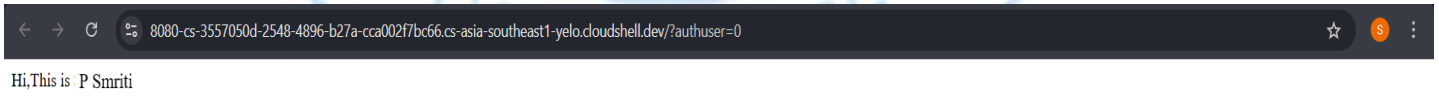
2. You can edit the code and configuration files as needed to develop your web application.

3. Test your application locally by clicking the "Run" button in the GAE Launcher.



Step 6: Deploy Your Web Application

Click the "Deploy" button in the GAE Launcher to upload your application to Google App Engine.



RESULT:

Thus the Google App Engine (GAE) launcher to launch web applications was successfully launched.